Advanced Dynamics — Equation Sheet

Kinematics · Newton's Laws · Energy & Momentum · Generalized Coordinates & Constraints · D'Alembert's Principle · Lagrangian & Hamiltonian Mechanics · Nonlinear Dynamics & Bifurcations · Nondimensionalization

Notation

- $N, A, B, \dots$ — reference frames (e.g. N = fixed/inertial)
- $\hat{n}_i, \hat{a}_i, \hat{b}_i$ — right-handed unit-vector basis fixed in N, A, B
- $\vec{r}_{P/O}$ — position of point P relative to point O; $\vec{r} \equiv {}^N\vec{r}$ unless noted otherwise
- ${}^N\vec{v}_P,\ {}^N\vec{a}_P$ — velocity/acceleration of P *as seen from* N
- $\vec{\omega}^{A/N}$ — angular velocity of frame A relative to N
- $\vec{\alpha}^{A/N} = {}^N\frac{d}{dt}\vec{\omega}^{A/N}$ — angular acceleration of A relative to N
- ${}^N\frac{d}{dt}(\cdot)$ — time derivative taken *by an observer fixed in* N
- $[R]^{B/A}$ — rotation (direction cosine) matrix: $\{\hat{b}\} = [R]^{B/A}\{\hat{a}\}$
- $q_1, \dots, q_N$ — generalized coordinates; $\vec{\gamma}_{ij} = \partial\vec{r}_i/\partial q_j$ — partial velocities
- $\vec{P} = m\vec{v}$ — linear momentum; $\vec{H}_O = \vec{r} \times \vec{P}$ — angular momentum about O
- $L = T - V$ — Lagrangian; $p_j = \partial L/\partial\dot{q}_j$ — conjugate (generalized) momentum
- $H = \sum_j p_j\dot{q}_j - L$ — Hamiltonian; λ_k — Lagrange multiplier for constraint ϕ_k

Rotations & Angular Velocity

Rotation matrices

Orthogonal, so $[R]^{-1} = [R]^T$; composing rotations through an intermediate frame A:

$$[R]^{B/N} = [R]^{B/A}[R]^{A/N}$$

Angular velocity addition (chain rule)

Valid for any chain of intermediate frames:

$$\vec{\omega}^{B/N} = \vec{\omega}^{B/A} + \vec{\omega}^{A/N}, \qquad \vec{\omega}^{C/N} = \vec{\omega}^{C/B} + \vec{\omega}^{B/A} + \vec{\omega}^{A/N}$$

Note the reversal/cancellation identities:

$$\vec{\omega}^{A/N} = -\vec{\omega}^{N/A}, \qquad \vec{\omega}^{A/A} = \vec{0}$$

Derivative of a rotating unit vector

If $\hat{b}_i$ is fixed in B (constant length/orientation relative to B):

$${}^N\frac{d}{dt}\hat{b}_i = \vec{\omega}^{B/N} \times \hat{b}_i$$

2D rotation example (N, P frames offset by α)

$$\hat{n}_1 = \hat{p}_1\cos\alpha + \hat{p}_2\sin\alpha, \qquad \hat{n}_2 = -\hat{p}_1\sin\alpha + \hat{p}_2\cos\alpha$$

$$\begin{Bmatrix}\hat{n}_1\\ \hat{n}_2\end{Bmatrix} = [R]^{N/P}\begin{Bmatrix}\hat{p}_1\\ \hat{p}_2\end{Bmatrix}, \qquad [R]^{N/P} = \begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}$$

(written ${}^N R_P$ in lecture)

$$[R]^{P/N} = \left([R]^{N/P}\right)^{-1} = \left([R]^{N/P}\right)^T = \begin{bmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{bmatrix}$$

Vector Differentiation Formula (VDF)

Also called the *transport theorem*: for *any* vector $\vec{q}$ (not necessarily fixed in either frame), relating the rate of change seen in N to the rate of change seen in a frame A rotating at $\vec{\omega}^{A/N}$:

$${}^N\frac{d}{dt}\vec{q} = {}^A\frac{d}{dt}\vec{q} + \vec{\omega}^{A/N} \times \vec{q}$$

Corollary: if $\vec{q}$ is fixed in A (i.e. ${}^A\frac{d}{dt}\vec{q} = 0$), this reduces to ${}^N\frac{d}{dt}\vec{q} = \vec{\omega}^{A/N} \times \vec{q}$ (previous box). This is the tool used to differentiate a vector expressed in a rotating basis one component at a time.

Position Vector Addition (Setup for Moving Frames)

For two points P and O' both located relative to a common origin O (equivalently $\vec{r} = \vec{R} + \vec{r}\,'$, with $\vec{R} = \vec{r}_{O'/O}$, $\vec{r}\,' = \vec{r}_{P/O'}$), position vectors add/subtract exactly like any vectors — the basic relation used to move between frames whose origins differ:

$$\vec{r}_{P/O} = \vec{r}_{P/O'} + \vec{r}_{O'/O} \qquad \Longleftrightarrow \qquad \vec{r}_{P/O'} = \vec{r}_{P/O} - \vec{r}_{O'/O}$$

Chained over several intermediate points: $\vec{r}_{P/O} = \vec{r}_{P/B} + \vec{r}_{B/A} + \vec{r}_{A/O}$. Applying the VDF to this relation term-by-term is exactly how the velocity/acceleration relations below are derived.

Velocity: Translating & Rotating Frames

Let O' be the (moving) origin of frame A, which translates *and* rotates ($\vec{\omega}^{A/N}$) relative to N, and let P be an arbitrary point:

$${}^N\vec{v}_P = {}^N\vec{v}_{O'} + {}^A\vec{v}_P + \vec{\omega}^{A/N} \times \vec{r}_{P/O'}$$

- ${}^N\vec{v}_{O'}$ — translational velocity of the moving origin
- ${}^A\vec{v}_P = {}^A\frac{d}{dt}\vec{r}_{P/O'}$ — velocity of P *relative to* A (as seen by an observer riding on A)
- $\vec{\omega}^{A/N} \times \vec{r}_{P/O'}$ — velocity contribution from A's rotation

Special cases

- Pure translation ($\vec{\omega}^{A/N} = 0$): ${}^N\vec{v}_P = {}^N\vec{v}_{O'} + {}^A\vec{v}_P$
- P fixed in A (${}^A\vec{v}_P = 0$, rigid-body point): ${}^N\vec{v}_P = {}^N\vec{v}_{O'} + \vec{\omega}^{A/N} \times \vec{r}_{P/O'}$
- Common origin ($O' = O$, ${}^N\vec{v}_{O'} = 0$): reduces to the transport theorem applied to $\vec{r}_{P/O}$

Acceleration: Translating & Rotating Frames

Differentiating the velocity relation again (the "five-term" / Coriolis theorem):

$$\begin{aligned}{}^N\vec{a}_P = {}^N\vec{a}_{O'} + {}^A\vec{a}_P + \vec{\alpha}^{A/N} \times \vec{r}_{P/O'}\\ + \vec{\omega}^{A/N} \times \left(\vec{\omega}^{A/N} \times \vec{r}_{P/O'}\right) + 2\,\vec{\omega}^{A/N} \times {}^A\vec{v}_P\end{aligned}$$

- ${}^N\vec{a}_{O'}$ — translational acceleration of the moving origin
- ${}^A\vec{a}_P = {}^A\frac{d}{dt}{}^A\vec{v}_P$ — acceleration of P relative to A
- $\vec{\alpha}^{A/N} \times \vec{r}_{P/O'}$ — Euler (angular acceleration) term
- $\vec{\omega}^{A/N} \times (\vec{\omega}^{A/N} \times \vec{r}_{P/O'})$ — centripetal term
- $2\,\vec{\omega}^{A/N} \times {}^A\vec{v}_P$ — Coriolis term

Special cases

- Pure translation ($\vec{\omega}^{A/N} = 0$): ${}^N\vec{a}_P = {}^N\vec{a}_{O'} + {}^A\vec{a}_P$
- P fixed in A (${}^A\vec{v}_P = {}^A\vec{a}_P = 0$, rigid-body point):

$${}^N\vec{a}_P = {}^N\vec{a}_{O'} + \vec{\alpha}^{A/N} \times \vec{r}_{P/O'} + \vec{\omega}^{A/N} \times (\vec{\omega}^{A/N} \times \vec{r}_{P/O'})$$

- P moving in A but O' fixed in N (common origin, no translation): drop the ${}^N\vec{a}_{O'}$ term — this is the form used for rotating-basis problems (e.g. spherical pendulum, rotating tube)

Newton's Laws of Motion

1st: $\sum\vec{F} = 0 \Rightarrow {}^N\dot{\vec{r}} = \text{const}$ **2nd:** $\sum\vec{F} = m\,{}^N\ddot{\vec{r}}$

3rd: $\vec{F}_{12} = -\vec{F}_{21}$ (action–reaction pairs)

Newton's 2nd law **requires** the acceleration relative to an inertial (Newtonian) frame N — never a rotating/accelerating frame directly.

Work & Energy

Kinetic & potential energy

$$T = \tfrac{1}{2}m|\dot{\vec{r}}|^2 = \tfrac{1}{2}m\,\dot{\vec{r}}\cdot\dot{\vec{r}}, \qquad V = V(\vec{r})$$

$$\vec{F} = -\nabla V(\vec{r}) \quad \text{(conservative/"potential" force)}$$

Common potentials

- Uniform gravity: $V = mgh$, $\vec{F} = -mg\,\hat{n}_3$
- Newtonian gravity: $V = -\dfrac{GmM}{r}$, $\vec{F} = \dfrac{GmM}{r^2}\hat{b}_1$ ($\hat{b}_1$: unit vector toward the attracting mass)
- Linear spring (stretch x): $V = \frac{1}{2}kx^2$, $\vec{F} = -kx\,\hat{n}_1$

Work–energy theorem

$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F}\cdot d\vec{r}, \qquad T_2 - T_1 = W$$

All forces conservative ($\vec{F} = -\nabla V$):

$$T_1 + V_1 = T_2 + V_2 \quad \text{(energy conserved)}$$

Only holds when *every* force doing work is conservative — friction or other non-conservative applied forces break it.

Momentum

Linear momentum

$$\vec{P} = m\dot{\vec{r}}, \qquad \dot{\vec{P}} = m\ddot{\vec{r}} = \sum \vec{F}$$

If $\sum \vec{F} = \vec{0}$, $\vec{P}$ is conserved.

Angular momentum (about inertial fixed point O)

$$\vec{H}_O = \vec{r}\times\vec{P}, \qquad \dot{\vec{H}}_O = \vec{r}\times\sum\vec{F} = \sum\vec{M}_O$$

"Rotational Newton's 2nd law." If $\sum \vec{M}_O = \vec{0}$, $\vec{H}_O$ is conserved.

Generalized Coordinates

- **Generalized coordinates** $q_1,\ldots,q_N$: minimal (or redundant) geometric variables (angles/distances) describing the configuration of a system
- **Configuration space:** the set of all configurations the system can take on, parametrized by the q_j
- A system may be described by a **redundant set** (more coordinates than DOF, related by constraint equations) or a **minimal set** (one coordinate per DOF, no constraints left)

$$n_{\text{DOF}} = \#(\text{gen. coords}) - \#(\text{indep. constraints})$$

Always reduce to the minimal set before deriving equations of motion.

Holonomic Constraints

Position-based: with N generalized coordinates and M constraints ($j = 1,\ldots,M$):

$$\phi_j(q_1,\ldots,q_N) = \phi_j(\vec{q}) = 0, \quad \text{or, if time-dependent,} \quad \phi_j(\vec{q},t) = 0$$

- **Scleronomic:** no explicit t — time-independent constraint
- **Rheonomic:** explicit t dependence — e.g. a prescribed moving support
- Each independent holonomic constraint removes exactly one DOF (box above)
- **Always rewritable as non-holonomic:** since $\phi_j \equiv 0 \Rightarrow \dot{\phi}_j \equiv 0$,

$$\dot{\phi}_j(\vec{q},t) = \sum_{i=1}^{N} \frac{\partial\phi_j}{\partial q_i}\dot{q}_i + \frac{\partial\phi_j}{\partial t} = 0 \qquad \left(a_{ji} \equiv \partial\phi_j/\partial q_i\right)$$

Non-Holonomic Constraints

Velocity-based: $\phi_j(\vec{q},\dot{\vec{q}},t) = 0$, usually taken *linear* in the velocities (Pfaffian form):

$$\phi_j = \sum_{i=1}^{N} a_{ji}(\vec{q},t)\,\dot{q}_i + a_{jt}(\vec{q},t) = 0$$

(no explicit t: $\phi_j = \sum_i a_{ji}(\vec{q})\dot{q}_i = 0$)

- Every holonomic constraint can be written this way (box above); the **converse is false** — a Pfaffian constraint is genuinely non-holonomic unless it is exact/integrable (or made exact with an integrating factor; check via the Frobenius condition)
- Non-holonomic constraints reduce the number of independent *velocities* but **not** the dimension of the configuration space
- **Example — knife-edge/skate/wheel constraint:** contact-point velocity must be perpendicular to the blade, $\vec{v}_A\cdot\hat{n} = 0$ with $\hat{n} = (-\sin\theta, \cos\theta)$:

$$\phi = -\sin\theta\,\dot{x} + \cos\theta\,\dot{y} = 0$$

With $(q_1,q_2,q_3) = (x,y,\theta)$: $a_1 = -\sin q_3$, $a_2 = \cos q_3$, $a_3 = 0$. Cannot be integrated to $\phi(\vec{q}) = 0$ — genuinely non-holonomic.

D'Alembert's Principle

Partial velocities

For particle i with position $\vec{r}_i(\vec{q},t)$, the partial velocity w.r.t. generalized speed $\dot{q}_j$:

$$\vec{\gamma}_{ij} = \frac{\partial\vec{r}_i}{\partial q_j} = \frac{\partial\dot{\vec{r}}_i}{\partial\dot{q}_j}$$

(equal because $\dot{\vec{r}}_i$ is linear in the $\dot{q}_j$'s — fastest computed by differentiating the already-known $\dot{\vec{r}}_i$ directly w.r.t. $\dot{q}_j$.)

Equations of motion

For a minimal set $q_1,\ldots,q_M$ and particles $i = 1,\ldots,n$:

$$\sum_{i=1}^{n}\left(\vec{F}_{Ai} - m_i\ddot{\vec{r}}_i\right)\cdot\vec{\gamma}_{ij} = 0, \qquad j = 1,\ldots,M$$

- $\vec{F}_{Ai}$ — *applied* (impressed) forces on particle i only (gravity, springs, motors, ...)
- **Ideal constraint forces are excluded** (rod tension, normal forces, frictionless-contact forces) — they do no virtual work consistent with the constraints and drop out automatically
- One scalar equation per generalized coordinate — exactly M equations for M DOF
- Equivalently, with generalized force $Q_j = \sum_i \vec{F}_{Ai}\cdot\vec{\gamma}_{ij}$: $\sum_i m_i\ddot{\vec{r}}_i\cdot\vec{\gamma}_{ij} = Q_j$

Recipe

1. Choose a minimal set $q_1,\ldots,q_M$ consistent with all holonomic constraints
2. Write $\vec{r}_i(\vec{q},t)$ for every particle; differentiate for $\dot{\vec{r}}_i, \ddot{\vec{r}}_i$
3. Compute $\vec{\gamma}_{ij} = \partial\dot{\vec{r}}_i/\partial\dot{q}_j$ for every particle/coordinate pair
4. Identify applied forces $\vec{F}_{Ai}$ on each particle (*exclude* constraint forces)
5. For each j: $\sum_i(\vec{F}_{Ai} - m_i\ddot{\vec{r}}_i)\cdot\vec{\gamma}_{ij} = 0 \Rightarrow M$ equations of motion
6. For linear (spring/mass) systems, these assemble into $[M]\ddot{\vec{q}}+[K]\vec{q} = \vec{F}$

Lagrangian Mechanics

Euler–Lagrange equations

For a minimal set $q_1,\ldots,q_M$; equivalent to D'Alembert's principle above (substitute $\vec{\gamma}_{ij} = \partial\dot{\vec{r}}_i/\partial\dot{q}_j$ into $L = T - V$ and simplify):

$$\frac{d}{dt}\left(\frac{\partial L}{\partial\dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = Q_j^{nc}, \qquad j = 1,\ldots,M$$

- Q_j^{nc} — generalized force from any applied force *not* already folded into V (a prescribed drive $F(t)$, friction, ...); $= 0$ if every applied force is conservative
- One 2nd-order ODE per DOF; ideal constraint forces are still automatically excluded, since $L(\vec{q},\dot{\vec{q}},t)$ is built from the minimal coordinates
- Cross terms quadratic in velocity (Coriolis-type, from $\partial T/\partial q_j$) fall out of $\frac{d}{dt}(\partial L/\partial\dot{q}_j) - \partial L/\partial q_j$ automatically — no need to track them separately as in Newtonian/D'Alembert bookkeeping

Matrix form & small oscillations

Multi-DOF equations of motion assemble into

$$M(\vec{q})\ddot{\vec{q}} + \vec{C}(\vec{q},\dot{\vec{q}}) + \vec{K}(\vec{q},t) = \vec{0}$$

(M: symmetric positive-definite mass/inertia matrix; $\vec{C}$: Coriolis/centripetal-type $\dot{q}^2$ terms; $\vec{K}$: stiffness/gravity/forcing terms). Linearizing about an equilibrium ($\sin(\cdot)\to(\cdot)$, $\cos(\cdot)\to 1$, drop $\dot{q}^2$ terms):

$$M\ddot{\vec{q}} + K\vec{q} = \vec{0}$$

Assume $\vec{q} = \vec{Q}e^{i\omega t}$: non-trivial solutions require the **characteristic equation**

$$\det\left(K - \omega^2 M\right) = 0 \quad \Rightarrow \quad \text{natural frequencies } \omega_i$$

Mode shape $\vec{Q}_i$ (up to overall scale) solves $(K - \omega_i^2 M)\vec{Q}_i = \vec{0}$; lowest ω_i = coordinates in phase, higher ω_i = out of phase.

Ignorable (cyclic) coordinates & Routhian reduction

If L has no explicit q_k-dependence ($\partial L/\partial q_k = 0$; only $\dot{q}_k$ appears), Lagrange's equation for it collapses to a conservation law:

$$\frac{\partial L}{\partial\dot{q}_k} = B_k = \text{const} \qquad \text{(conserved generalized momentum)}$$

Invert for $\dot{q}_k = f(\vec{q}_{\text{rest}}, B_k)$ and use it to eliminate $\dot{q}_k$ from the remaining dynamics via the **Routhian**

$$R(\vec{q}_{\text{rest}}, \dot{\vec{q}}_{\text{rest}}; B_k) \equiv L - \dot{q}_k B_k \Big|_{\dot{q}_k \to f(\vec{q}_{\text{rest}}, B_k)}$$

$$\frac{d}{dt}\frac{\partial R}{\partial \dot{q}_j} - \frac{\partial R}{\partial q_j} = 0 \quad (j \neq k)$$

This reduces the problem by one coordinate; recover $q_k(t)$ afterward by integrating $\dot{q}_k = f(\vec{q}_{\text{rest}}(t), B_k)$.

Constraint forces via Lagrange multipliers

For M coordinates with m Pfaffian constraints $\phi_k = \sum_i a_{ki}\dot{q}_i + a_{kt} = 0$ ($k = 1, \ldots, m$; a holonomic constraint left un-eliminated works the same way, with $a_{ki} = \partial\phi_k/\partial q_i$):

$$\boxed{\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = \sum_{k=1}^{m} \lambda_k\, a_{kj}, \qquad j = 1, \ldots, M}$$

- M equations of motion + m constraint equations $\phi_k = 0$ close the system for $\vec{q}(t)$ **and** $\lambda_k(t)$
- $\sum_k \lambda_k a_{kj}$ is exactly the constraint force conjugate to q_j — this is how the forces D'Alembert/plain Lagrange discard get recovered (e.g. rod tension, knife-edge normal force)
- **Recipe:** write $L = T - V$; list a_{ki} from each ϕ_k; write the M boxed equations plus the m constraints $\phi_k = 0$; solve algebraically for λ_k (introducing a body-frame speed like $u \equiv \dot{\vec{r}} \cdot \hat{b}_1$ often decouples the algebra)

Hamiltonian Mechanics

Generalized momentum & the Hamiltonian

$$p_j \equiv \frac{\partial L}{\partial \dot{q}_j}$$

$$\boxed{H(\vec{q}, \vec{p}, t) \equiv \sum_j p_j \dot{q}_j - L \qquad \text{(Legendre transform of } L\text{)}}$$

Invert $p_j(\vec{q}, \dot{\vec{q}}, t)$ for $\dot{q}_j(\vec{q}, \vec{p}, t)$ and substitute so H is expressed purely in $(\vec{q}, \vec{p}, t)$.

When does $H = T + V$?

Split T by degree in the $\dot{q}_j$: $T = T_2 + T_1 + T_0$ (T_2 homogeneous quadratic, T_1 linear, T_0 velocity-independent — all can depend on $\vec{q}, t$). In general

$$\boxed{H = T_2 - T_0 + V \qquad \text{(the } \textit{Jacobi integral}\text{)}}$$

- Reduces to $H = T + V$ **only** when $T_1 = T_0 = 0$: $\vec{r}_i(\vec{q})$ has no explicit t-dependence (scleronomic transformation) so $T = T_2$ is homogeneous quadratic in $\dot{\vec{q}}$
- Rheonomic transformations (a prescribed $\theta(t) = \omega_0 t$, a moving support, …) generally give $T_1, T_0 \neq 0$, so $H \neq T + V$ even when H is conserved
- H is conserved whenever $\partial L/\partial t = 0$ (no explicit t left *after* substituting any prescribed motion), regardless of whether $H = T + V$

Hamilton's canonical equations

$$\boxed{\dot{q}_j = \frac{\partial H}{\partial p_j}, \qquad \dot{p}_j = -\frac{\partial H}{\partial q_j}}$$

$2M$ first-order ODEs replace the M second-order Euler–Lagrange equations — the natural form for numerical IVP integration. If q_k is cyclic ($\partial H/\partial q_k = 0$): $\dot{p}_k = 0 \Rightarrow p_k = \text{const}$ (matches the Lagrangian/Routhian result, $p_k \equiv B_k$).

Canonical transformations & generating functions

$(\vec{q}, \vec{p}) \to (\vec{Q}, \vec{P})$ is **canonical** if Hamilton's equations hold in the new variables for some $H'(\vec{Q}, \vec{P}, t)$. A **type-1 generating function** $F(\vec{q}, \vec{Q}, t)$ produces one via

$$\boxed{p_j = \frac{\partial F}{\partial q_j}, \qquad P_j = -\frac{\partial F}{\partial Q_j}, \qquad H' = H + \frac{\partial F}{\partial t}}$$

- If F has no explicit t: $H'(\vec{Q}, \vec{P})$ is just H re-expressed in the new variables
- **Goal:** pick F so H' is simple (e.g. every Q_j cyclic) — then $\dot{P}_j = 0$ and $\dot{Q}_j = \partial H'/\partial P_j = \text{const}$, trivially integrable
- **Recipe:** (1) $p(q, Q) = \partial F/\partial q$; (2) $P(q, Q) = -\partial F/\partial Q$; (3) invert for $q(Q, P)$, then $p(Q, P)$; (4) substitute into $H(q, p)$ to get $H'(Q, P)$; (5) solve the (now trivial) EOM in (Q, P); (6) transform back for $q(t), p(t)$

1-D Dynamical Systems & Bifurcations

Fixed points & linear stability

System $\dot{x} = f(x; r)$, r a control (parameter).

$$\boxed{\text{Fixed pts: } f(x^*; r) = 0 \qquad \text{Stability: } \dot{\eta} = f'(x^*)\,\eta, \quad \eta \equiv x - x^*}$$

- $f'(x^*) < 0$: stable/attracting $\quad f'(x^*) > 0$: unstable/repelling
- $f'(x^*) = 0$: linearization inconclusive — the *marginal/degenerate* case where bifurcations live

Recipe: locate a bifurcation

1. Find all fixed points $x^*(r)$: solve $f(x; r) = 0$
2. Compute $f_x(x; r) = \partial f/\partial x$ and evaluate along each branch $x^*(r)$
3. A bifurcation sits at (x_c, r_c) where the branch is *degenerate*:
   $$f(x_c; r_c) = 0 \quad \text{and} \quad f_x(x_c; r_c) = 0$$
   (solve these two equations simultaneously for x_c, r_c — this is exactly where two branches of $x^*(r)$ meet, or a branch's stability flips)
4. Classify using the recipe/table below (Taylor-expand f about (x_c, r_c) if it's not already in normal form)

Fast decision path (given explicit $f(x, r)$)

1. Is $x = x_0$ a root of f for *every* r (a "trivial" branch, often $x_0 = 0$)?
   No → solve $f = 0$, $f_x = 0$ together for isolated (x_c, r_c) → **saddle-node**.
   Yes → continue.
2. Is $f(x, r)$ odd about x_0, i.e. $f(x_0 - x, r) = -f(x_0 + x, r)$?
   No → **transcritical** at r_c solving $f_x(x_0, r_c) = 0$.
   Yes → **pitchfork** at that r_c; Taylor/series-expand $f \approx (r - r_c)(x - x_0) + a\,(x - x_0)^3$ (expand any sinh, $\frac{x}{1+x^2}$, etc. to cubic order) — $a < 0$ supercritical, $a > 0$ subcritical.

Classification (normal forms)

Let $\xi = x - x_c$, $\mu = r - r_c$.

- **Saddle-node (fold):** $\dot{\xi} = \mu - \xi^2$ (or $\mu + \xi^2$) — *no* fixed point at $\xi = 0$ for all μ; two fixed points ($x^* \propto \pm\sqrt{\mu}$) collide and annihilate as μ crosses 0, existing only on one side of r_c. Test: $f_r \neq 0$, $f_{xx} \neq 0$.
- **Transcritical:** $\dot{\xi} = \mu\xi - \xi^2$ — $\xi = 0$ *is* a fixed point for *all* μ (trivial branch persists forever); a second branch $\xi = \mu$ crosses it at $\mu = 0$ and the two *exchange stability*; total # of fixed points is unchanged across r_c. Test: $f_{x\mu} \neq 0$, $f_{xx} \neq 0$.
- **Supercritical pitchfork:** $\dot{\xi} = \mu\xi - \xi^3$ — symmetric ($f(-\xi) = -f(\xi)$); trivial branch $\xi = 0$ stable for $\mu < 0$, goes unstable at $\mu = 0$ and sheds *two new stable* branches $\xi = \pm\sqrt{\mu}$ for $\mu > 0$. Cubic coefficient < 0.
- **Subcritical pitchfork:** $\dot{\xi} = \mu\xi + \xi^3$ — same symmetry, but sheds *two unstable* branches for $\mu < 0$; trivial branch is unstable for $\mu > 0$ with no nearby stable state (real systems jump/hysterese to a distant attractor via a higher-order saddle-node). Cubic coefficient > 0.

Phase line (1-D phase portrait)

Plot $\dot{x}$ vs. x for fixed r: mark equilibria where the curve crosses $\dot{x} = 0$, arrows on the x-axis between crossings showing the sign of $\dot{x}$ (right if $f > 0$, left if $f < 0$).

- Filled dot: stable (f goes $+ \to -$ through it, arrows point *in*); open dot (∘): unstable ($- \to +$, arrows point *out*)
- Half-filled dot: semi-stable (same sign on both sides — the signature of a saddle-node exactly at $r = r_c$)
- Sweeping r and stacking phase lines side-by-side (equilibria vs. r) reconstructs the bifurcation diagram; solid branch = stable, dashed = unstable

(2-D extension — nullclines, Jacobian trace/det, Hopf bifurcation: TBD)

Nondimensionalizing an Equation of Motion

Recipe

1. Write the dimensional EOM and list every dimensional quantity appearing in it (state variable(s), t, and all parameters: $m, k, b, g, \ell, \ldots$)
2. Choose a characteristic **length** L and **time** T built from the problem's own parameters (e.g. L = a natural length/amplitude in the geometry, $T = \sqrt{m/k}$ or $1/\omega_n$) — these are what get scaled out
3. Define dimensionless variables $\xi = x/L$, $\tau = t/T$ (one pair per independent physical dimension in play; add a scaled force/energy if one

appears)

4. Convert derivatives by chain rule: $\dot{x} = \dfrac{L}{T}\xi'$, $\ddot{x} = \dfrac{L}{T^2}\xi''$ (primes $= d/d\tau$)
5. Substitute into the EOM; divide the whole equation by the coefficient of the *highest* derivative so it has unit leading coefficient
6. Collect every remaining combination of parameters into as few dimensionless groups as possible (Buckingham π: #groups = #parameters − #independent dimensions); name each one ($\zeta, \epsilon, \dots$) — these are the only "knobs" left in the problem
7. Sanity check: every leftover coefficient multiplying a ξ, ξ', ξ'' term must be a pure number (no residual kg/m/s)

Worked template (spring–mass–damper type)

Dimensional: $m\ddot{x} + b\dot{x} + kx = F(t)$ (e.g. a gravity component or drive). With $T = \sqrt{m/k}$, $L =$ a natural length scale, $\tau = t/T$, $\xi = x/L$:

$$\xi'' + 2\zeta\,\xi' + \xi = \tilde{F}(\tau), \qquad 2\zeta \equiv \frac{b}{\sqrt{mk}}, \qquad \tilde{F} \equiv \frac{F\,T^2}{mL}$$

ζ (damping ratio) is the one leftover dimensionless group here; a tilted/geometric system typically adds one more (an angle α — already dimensionless — or a length ratio a/L), matching "3 dimensionless parameters" style results.

Overdamped/singular limit ⇒ drops the order by one

If a derivative's dimensionless coefficient is formally small (heavy damping ⇒ small effective inertia, or a genuinely small mass), write it as $\epsilon\,\xi'' + \xi' + \cdots = 0$ with $\epsilon \ll 1$ and set $\epsilon \to 0$ (quasi-static/regular-perturbation assumption):

$$\epsilon\,\xi'' + \xi' = g(\xi; \vec{p}) \quad \xrightarrow{\epsilon \to 0} \quad \xi' = g(\xi; \vec{p})$$

This collapses a 2nd-order (2-D phase-plane) system to a **1-D, 1st-order** system — the phase-line/bifurcation analysis above applies directly, with the remaining dimensionless groups $\vec{p}$ playing the role of r.